Tristan Ryoma Fuchs

フックストリスタン龍馬

Master's student (M2)
Department of Physics, The University of Tokyo
Arita Group · MERIT-WINGS program student

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RESEARCH INTERESTS

Superconductivity · first-principles (DFT) calculations

EDUCATION

Apr 2025 – present
Master's Program in Physics, The University of Tokyo
Supervisor: Ryotaro Arita.

Apr 2021 – Mar 2025
B.Sc. in Physics, Department of Physics, Faculty of Science, Tohoku University
Supervisor: Takashi Koretsune.
First-principles Migdal–Eliashberg calculation of the superconducting gap functions of Al and H_3S.

Sep 2016 – Jun 2020
Shanghai Jincai High School, International Division, Shanghai, China
Grades 9–12; International Baccalaureate Diploma Programme (Grades 11–12).

Apr 2008 – Aug 2016
Shanghai Japanese School, Pudong Campus, Shanghai, China
Grades 1–9.

PUBLICATIONS

1. **T. R. Fuchs**, T. Nomoto, H. Watanabe, and R. Arita, "Microscopic calculation of coherence lengths and magnetic penetration depth in multiband superconductors," arXiv:2609.06014 [cond-mat.supr-con] (2026), under review.

PRESENTATIONS

International Conferences

Dec 2026
Microscopic Calculation of Characteristic Length Scales in Multi-Band Superconductors POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Quantum Materials Symposium (i-QMS 2026), Jeju, Korea

Nov 2026
Coherence Lengths and Magnetic Penetration Depth in Multiband Superconductors from Microscopic Calculations POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Workshop on Frontiers in Ab-initio Many-Body Interactions (FAMBI 2026), Sendai, Japan

Nov 2026
Coherence Lengths and Magnetic Penetration Depth in Multiband Superconductors Based on Microscopic Calculations POSTER SCHEDULED
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 27th Asian Workshop on First-Principles Electronic Structure Calculations (ASIAN-27), Kaohsiung, Taiwan

Sep 2026
Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Conference on Strongly Correlated Electron Systems (SCES 2026), Toyama, Japan

Sep 2025
Microscopic Calculation of Coherence Length and Magnetic Penetration Depth in Superconductors POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Autumn School on Correlated Electrons: Understanding Correlated Materials with DMFT, Forschungszentrum Jülich, Germany

2026
3 other poster presentations
CDS × QUAST Joint Workshop: Frontiers in Correlated Quantum Matter, The University of Tokyo (Nov 2026, scheduled) · International Conference on Asymmetric Quantum Matters (Asymmetry 2026), Toyonaka (Sep 2026) · CEMS Topical Meeting on Recent Trends in Superconductivity, RIKEN, Wako (Feb 2026)

Domestic Conferences (Japan)

Sep 2026 — **Calculation of Coherence Lengths and Magnetic Penetration Depth Based on the Real-Space Bogoliubov–de Gennes Method** POSTER
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 81st Annual Meeting of the Physical Society of Japan (JPS), The University of Tokyo, Komaba Campus (in Japanese)

Sep 2026 — **Microscopic Calculation of Characteristic Length Scales in Multi-band Superconductors Based on the Extended Ginzburg–Landau Method** POSTER BEST PRESENTATION AWARD
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — MERIT Camp 2026 (MERIT-WINGS, The University of Tokyo), Minano, Saitama (in Japanese)

Aug 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors Based on Extended Ginzburg–Landau Theory** ORAL
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — 71st Condensed Matter Physics Summer School (物性若手夏の学校), Aso, Kumamoto (in Japanese)

Mar 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors** ORAL
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — JPS 2026 Spring Meeting, online (in Japanese)

Mar 2026 — **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors** POSTER BEST POSTER AWARD
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Computational Physics Spring School 2026 (計算物理春の学校), Naha, Okinawa (in Japanese)

2026 — **3 other presentations (1 oral, 2 poster)**
71st Condensed Matter Physics Summer School (物性若手夏の学校), Aso (Aug 2026, poster) · Kickoff Meeting of the Transformative Research Area (A) "Asymmetric Quantum Matters," Nasu (May 2026, poster) · 15th MERIT Colloquium, The University of Tokyo (Jan 2026, oral) — all in Japanese

AWARDS AND FELLOWSHIPS

Sep 2026 — **Best Presentation Award, MERIT Camp 2026**

Mar 2026 — **Best Poster Award, Computational Physics Spring School 2026** (計算物理春の学校)

Oct 2025 – present — **MERIT-WINGS Program Student**
World-leading Innovative Graduate Study Program for Materials Research, Information, and Technology (MERIT-WINGS), The University of Tokyo

EXPERIENCE

Dec 2023 – present — **Student Staff Lead, Nagase Brothers Inc. (education company)**
Oct 2024 – present: development of internal automation tools at Yotsuya Otsuka
Dec 2023 – Mar 2026: project management at Toshin Digital University

Spring 2024 — **Teaching Assistant, 学問論 (first-year general education course), Tohoku University**

LEADERSHIP AND ACTIVITIES

Oct 2025 – Sep 2026 — **Deputy Organizer, MERIT Camp** — The University of Tokyo

Oct 2023 – Mar 2025 — **Student Staff, Kita-Aobayama Library (Science & Pharmacy Library)** — Tohoku University

Apr 2022 – Mar 2023 — **Vice President, 物理の会 ("Butsuri no Kai," physics seminar society)** — Tohoku University

CERTIFICATIONS AND TEST SCORES

2026 — **EIKEN Grade 1**
実用英語技能検定（英検）1級

2025 — **Applied Information Technology Engineer Examination**
応用情報技術者試験 — passed

2024 — **TOEFL iBT 105**

2024 — **TOEIC Listening & Reading 960**

2022 — **HSK Level 4**
汉语水平考试 (Chinese Proficiency Test)

LANGUAGES AND SKILLS

Languages: Japanese (native), English (fluent), German (conversational), Chinese (basic)

Programming: Julia (main language for research), Python, Google Apps Script

Last updated: October 2026